

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404707
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Russell; Don J. et al.

Systems and methods for preventing a walkway platform flip door from being raised without a guardrail installed

Abstract

In an example, a system for preventing a flip door from being raised without a guardrail installed is disclosed. The system includes a guardrail having a post, and the post having a post dog. The system also includes a safety latch mechanism configured to be coupled to a fixed structure. The safety latch mechanism includes a body, a socket disposed in the body and configured to receive the post, and a door latch rotatably coupled to the body and biased in a latched position that is configured to retain a flip door in a lowered position. Based on the post being inserted into the socket of the safety latch mechanism, the post dog engages with and rotates the door latch from the latched position to an unlatched position, thereby allowing the flip door to be raised from the lowered position to a raised position.

Inventors: Russell; Don J. (Arlington, WA), Dunlap; Brennan J. (Snohomish, WA)

Applicant: The Boeing Company (Chicago, IL)

Family ID: 80953654

Assignee: The Boeing Company (Arlington, VA)

Appl. No.: 17/369560

Filed: July 07, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230011346 A1	Jan. 12, 2023

Publication Classification

Int. Cl.: E05C3/06 (20060101); E04G5/00 (20060101); E04G5/14 (20060101); E05C3/00 (20060101)

U.S. Cl.:

CPC **E05C3/06** (20130101); **E04G5/001** (20130101); **E04G5/14** (20130101); **E05C3/004** (20130101);

Field of Classification Search

CPC: E04G (1/15); E04G (1/00); E04G (5/006); E04G (5/14); E04G (5/001); E04G (5/141); E04G (2001/157); E04G (21/3233); E04G (21/3228); E05C (3/14); E05C (3/006); E05C (3/16); E05C (3/162); E05C (3/36); B64F (5/10); B64F (5/40); B66F (11/04); B66F (17/006); E05B (63/00); E05B (63/0008); E05B (15/00); E05B (65/00); E04F (11/18); E04F (11/1865); E04F (2011/1868); E04H (17/14); E04H (17/20); E06B (3/38); E06B (11/00); E05Y (2201/434); B25H (1/20); B60R (3/005); E04B (57/00); E04B (35/008); E05D (11/1007); E05D (11/1014); Y10T (292/17)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5941024	12/1998	Journault	182/112	E02D 29/1463
6138812	12/1999	Kimener	49/133	E06B 11/00
7841645	12/2009	Diamond	296/165	B60P 3/34
8365472	12/2012	Grimaldi	114/364	B63B 17/04
8403109	12/2012	Bennett	182/113	E06C 5/02
8479884	12/2012	Mizell	182/141	E04G 3/00
9027712	12/2014	Melton	182/36	B65G 69/22
9409755	12/2015	Melton	N/A	B66F 11/04
10730372	12/2019	Roth	N/A	B60P 1/435
11156008	12/2020	Phan	N/A	B64F 5/60
11319713	12/2021	O'Keefe	N/A	B66F 11/04
11512477	12/2021	Restivo	N/A	E04G 5/14
11608649	12/2022	Hammond	N/A	E04G 5/001
11773606	12/2022	Phan	182/20	E04G 5/14
2003/0020253	12/2002	Albert Bosman	280/164.1	B60R 3/005
2006/0054392	12/2005	MacDonald	182/113	B66F 11/04
2013/0119651	12/2012	Conny	248/214	B60R 3/005
2015/0034418	12/2014	Melton	182/113	B66F 11/04
2018/0272951	12/2017	Oliver	N/A	B60R 3/005
2019/0135185	12/2018	Oliver	N/A	B60R 3/005
2019/0360219	12/2018	O'Keefe	N/A	E04G 3/28
2020/0165828	12/2019	Phan	N/A	E04G 21/3228
2021/0017774	12/2020	Phan	N/A	E04G 21/3228
2021/0087823	12/2020	Stawski	N/A	E04F 11/1865
2021/0156152	12/2020	Restivo	N/A	E04F 11/18

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201793023	12/2010	CN	N/A
109733637	12/2018	CN	N/A
111216871	12/2019	CN	B64C 1/1423
111395722	12/2019	CN	E04G 1/15
111498765	12/2019	CN	N/A
112210393	12/2020	CN	C10B 41/00
114772515	12/2021	CN	N/A
115419318	12/2021	CN	N/A
116464333	12/2022	CN	N/A
116905780	12/2022	CN	N/A
2572759	12/2012	EP	A62B 99/00
3 878 757	12/2020	EP	N/A
2980512	12/2012	FR	E06B 11/00
20140068451	12/2013	KR	N/A
1020140068451	12/2013	KR	N/A
20190017837	12/2018	KR	N/A
1020190017837	12/2018	KR	N/A

OTHER PUBLICATIONS

Global Dossier Jan. 23, 2025. cited by examiner

Extended European Search Report prepared by the European Patent Office in application No. EP 22 16 4657.3 dated Sep. 19, 2022. cited by applicant

Primary Examiner: Mattei; Brian D

Assistant Examiner: McFarland; Kathleen M.

Attorney, Agent or Firm: Walters & Wasylyna LLC

Background/Summary

FIELD

(1) The present disclosure generally relates to systems and methods involving adjustable walkway platforms and guardrails, and more particularly, to preventing an adjustable walkway platform flip door from being raised without having a guardrail installed.

BACKGROUND

(2) During their manufacture in an airplane production facility, aircraft move along the production line between different stages of assembly. This periodic (e.g., daily) movement is called “pulsing” the production line.

(3) In some stages of assembly, personnel may access elevated portions of the aircraft by way of fixed deck platforms positioned within a few feet from the respective sides of the aircraft's fuselage. In such scenarios, movable deck platforms (e.g., flip doors) may be coupled to the fixed deck platforms and may be operable to provide personnel with the remaining platform access up to the body of the aircraft.

(4) Prior to pulsing the production line, the movable deck platforms may be manually rotated (e.g., raised upward or lowered downward away from the fuselage) so as to provide sufficient clearance

for airplane movement along the production line. To avoid falls from the fixed deck platform at this stage, removable guardrails can be manually installed near the end of the fixed deck platform.

(5) Once a new airplane is in a proper position along the production line, the guardrails may be manually removed and the movable deck platform could be reinstalled or rotated to a horizontal position to once again close the gap between the fixed deck platform and the aircraft fuselage and provide standing access for personnel immediately next to the fuselage.

(6) Existing designs for movable deck platforms typically rely on manual installation of guardrails or other forms of fall protection around the deck platforms after the deck platforms have been raised. However, with such designs, personnel often forget to install the guardrails, which can put the personnel at risk of falling. Thus, a safer alternative design is desired.

SUMMARY

(7) In an example, a system for preventing a flip door from being raised without a guardrail installed is described. The system comprises a guardrail comprising a post, the post having a post dog. The system also comprises a safety latch mechanism configured to be coupled to a fixed structure. The safety latch mechanism comprises a body, a socket disposed in the body and configured to receive the post, and a door latch rotatably coupled to the body and biased in a latched position that is configured to retain a flip door in a lowered position. Based on the post being inserted into the socket of the safety latch mechanism, the post dog engages with and rotates the door latch from the latched position to an unlatched position, thereby allowing the flip door to be raised from the lowered position to a raised position.

(8) In another example, another system for preventing a flip door from being raised without a guardrail installed is described. The system comprises a guardrail comprising two posts, each post having a respective post dog. The system also comprises two safety latch mechanisms configured to be coupled to a fixed structure on opposing sides of a flip door. Each safety latch mechanism comprises a body, a socket disposed in the body and configured to receive a respective post of the two posts, and a door latch rotatably coupled to the body and biased in a latched position that retains the flip door in a lowered position. Based on the respective post being inserted into the socket of the safety latch mechanism, the respective post dog engages with and rotates the door latch from the latched position to an unlatched position, thereby allowing the flip door to be raised from the lowered position to a raised position.

(9) In another example, a method is described. The method comprises positioning each of two posts of a guardrail proximate to a respective socket disposed in a respective one of two safety latch mechanisms coupled to a fixed structure on opposing sides of a flip door, each post having a respective post dog and each safety latch mechanism comprising a respective body having the respective socket disposed therein, a respective door latch rotatably coupled to the body and biased in a latched position that retains the flip door in a lowered position, and a respective post latch that is rotatably coupled to the body, biased in a locked position, and retained in an unlocked position via engagement with the flip door based on the flip door being in the lowered position. The method also comprises inserting each respective post of the two posts into the respective socket such that the respective post dog engages with and rotates the respective door latch from the latched position to an unlatched position, thereby allowing the flip door to be raised from the lowered position to a raised position. The method also comprises raising the flip door from the lowered position to the raised position, thereby disengaging the flip door from the respective post latch, whereupon the respective post latch rotates from the unlocked position to the locked position, engages with the respective post dog, and retains the respective post in the safety latch mechanism.

(10) The features, functions, and advantages that have been discussed can be achieved independently in various examples or may be combined in yet other examples. Further details of the examples can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) The novel features believed characteristic of the illustrative examples are set forth in the appended claims. The illustrative examples, however, as well as a preferred mode of use, further objectives and descriptions thereof, will best be understood by reference to the following detailed description of an illustrative example of the present disclosure when read in conjunction with the accompanying drawings, wherein:
- (2) FIG. 1 depicts a system, according to an example implementation.
- (3) FIG. 2 depicts the safety latch mechanism of FIG. 1, according to an example implementation.
- (4) FIG. 3 depicts the guardrail of FIG. 1, according to an example implementation.
- (5) FIG. 4 depicts the guardrail of FIG. 1, according to an example implementation.
- (6) FIG. 5 depicts an operating scenario involving the system of FIG. 1, according to an example implementation.
- (7) FIG. 6 depicts an operating scenario involving the system of FIG. 1, according to an example implementation.
- (8) FIG. 7 depicts an operating scenario involving the system of FIG. 1, according to an example implementation.
- (9) FIG. 8 depicts an operating scenario involving the system of FIG. 1, according to an example implementation.
- (10) FIG. 9 depicts an operating scenario involving the system of FIG. 1, according to an example implementation.
- (11) FIG. 10 depicts an operating scenario involving the system of FIG. 1, according to an example implementation.
- (12) FIG. 11 depicts an operating scenario involving the system of FIG. 1, according to an example implementation.
- (13) FIG. 12 depicts an operating scenario involving the system of FIG. 1, according to an example implementation.
- (14) FIG. 13 depicts an operating scenario involving the system of FIG. 1, according to an example implementation.
- (15) FIG. 14 depicts an operating scenario involving the system of FIG. 1, according to an example implementation.
- (16) FIG. 15 depicts an operating scenario involving the system of FIG. 1, according to an example implementation.
- (17) FIG. 16 shows a flowchart of an example method, according to an example implementation.

DETAILED DESCRIPTION

- (18) Disclosed examples will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all of the disclosed examples are shown. Indeed, several different examples may be described and should not be construed as limited to the examples set forth herein. Rather, these examples are described so that this disclosure will be thorough and complete and will fully convey the scope of the disclosure to those skilled in the art.
- (19) By the terms “substantially,” “about,” “approximately,” and “proximate” used herein, it is meant that the recited characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known to skill in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide.
- (20) Unless otherwise specifically noted, elements depicted in the drawings are not necessarily drawn to scale.
- (21) Within examples, described herein is a safety latch mechanism for preventing a movable deck

platform (e.g., a flip door) from being raised without a guardrail (or guardrails) installed. To facilitate this, the safety latch mechanism includes a body, a socket for receiving a post of the guardrail, and at least one door latch rotatably coupled to the body and biased (e.g., spring biased) in a latched position that is configured to retain the flip door in a lowered position. When the post of the guardrail is then inserted into the socket, a post dog of the post engages with the at least one door latch and rotates the at least one door latch to an unlatched position, thus allowing the flip door to be raised. Within examples, the safety latch mechanism can be installed on a fixed structure in an outboard fashion (e.g., coupled to an exterior of the fixed structure to which the flip door is coupled, as opposed to an underside or interior of the fixed structure).

(22) The safety latch mechanism can also include at least one post latch that is biased in a locked position, but is retained in an unlocked position via engagement with a dog on the flip door when the flip door is in the lowered position. Thus, when the flip door is raised, the dog disengages from the at least one post latch, thus causing the at least one post latch to rotate, engage with the post dog of the post, and lock the post in the safety latch mechanism until the flip door is lowered back to the lowered position.

(23) With the arrangement described above, safety is increased since the flip door cannot be raised until a protective guardrail is installed.

(24) These and other improvements are described in more detail below. Implementations described below are for purposes of example. The implementations described below, as well as other implementations, may provide other improvements as well.

(25) Referring now to the figures, FIG. 1 illustrates a system **100**, according to an example implementation. The various elements of system **100** could be formed from one or more materials such as aluminum, steel, plastic, and/or another material. Furthermore, it will be understood that such elements could additionally or alternatively be formed from machined parts.

(26) The system **100** includes a guardrail **102**, which includes a post **104** having a post dog **106**. In some embodiments, the guardrail **102** can include other parts as well that are not explicitly shown in FIG. 1, such as at least one rail (e.g., a top rail and a base rail, or a top rail and one or more extending rails perpendicular to the top rail) and at least one additional post. In such scenarios, the post **104** and the at least one additional post could be coupled to the top rail and the base rail.

(27) The system **100** also includes a safety latch mechanism **108** configured to be coupled to a fixed structure **110**. For example, the safety latch mechanism **108** can be coupled to the fixed structure **110** by way of one or more fasteners (e.g., screws, bolts, etc.). The fixed structure **110** can be a deck structure having one or more deck platforms on which personnel can walk and stand.

(28) The safety latch mechanism **108** includes a body **112**, a socket **114** disposed in the body **112** and configured to receive the post **104**, and a door latch **116**. The door latch **116** is rotatably coupled to the body **112** and biased in a latched position that is configured to retain a flip door **118** in a lowered position. Within examples, the door latch **116** can be a spring-loaded latch that is biased in the latched position by way of one or more springs. In alternative embodiments, the door latch **116** can be biased by way of some other mechanism, other than a spring.

(29) As described herein, elements that are “rotatably coupled” to one another could be coupled by way of a pin, a rotary bearing; or at least one rotary bushing and a stripper bolt. Other types of flexible and/or adjustable couplings are contemplated to join the relevant elements in a rotatable/pivotable fashion. For example, it will be understood that other types of rotatable couplings are contemplated herein. Without limitation, elements of the system **100** could be rotatably coupled to one another by way of a straight or articulated axle, a clevis pin, a plain bearing (e.g., sleeve bearing/bushing), a ball bearing, or roller bearing, among other possibilities.

(30) The flip door **118** can be rotatably coupled to the fixed structure **110** such that the flip door **118** can rotate between the lowered position and a raised position. To unlatch the door latch **116** and thus allow the flip door **118** to be raised from the lowered position to the raised position, the guardrail **102** is inserted into the safety latch mechanism **108**. In particular, based on the post **104**

being inserted into the socket **114** of the safety latch mechanism **108**, the post dog **106** engages with and rotates the door latch **116** from the latched position to an unlatched position, thereby allowing the flip door **118** to be raised from the lowered position to the raised position.

Furthermore, based on the flip door **118** being lowered to the lowered position and the post **104** being removed from the socket **114**, the post dog **106** can disengage from the door latch **116**, thus rotating the door latch **116** to the latched position at which the door latch **116** retains the flip door **118** in the lowered position.

(31) In some embodiments, the safety latch mechanism **108** also includes a post latch **120** rotatably coupled to the body **112** and biased in a locked position. In such a scenario, the flip door **118** includes a dog **122** that is configured to engage with and retain the post latch **120** in an unlocked position based on the flip door **118** being in the lowered position. Within examples, the post latch **120** can be a spring-loaded latch that is biased in the locked position by way of one or more springs. In alternative embodiments, the post latch **120** can be biased by way of some other mechanism, other than a spring.

(32) The post latch **120** can also be involved in helping promote engagement between the guardrail **102** and the safety latch mechanism **108**. In particular, based on the post **104** being inserted into the socket **114** of the safety latch mechanism **108** and the flip door **118** being raised to the raised position, the dog **122** is configured to disengage from the post latch **120**, thereby causing the post latch **120** to rotate from the unlocked position to the locked position, whereupon the post latch **120** engages with the post dog **106** and retains the post **104** in the safety latch mechanism **108**. Additionally, the post latch **120** can be involved in disengaging the guardrail **102** from the safety latch mechanism **108**. In particular, based on the flip door **118** being lowered from the raised position to the lowered position, the dog **122** is configured to engage with the post latch **120**, causing the post latch **120** to rotate from the locked position to the unlocked position and disengage from the post dog **106**, thereby allowing the post **104** to be removed from the socket **114**. Further, based on the post **104** being removed from the socket **114**, the door latch **116** rotates from the unlatched position to the latched position, thereby retaining the flip door **118** in the lowered position.

(33) The post dog **106** can be configured to facilitate engagement with the door latch **116** and the post latch **120**. For instance, in some embodiments, the post dog **106** has a tapered portion **124** and a recessed portion **126**.

(34) The tapered portion **124** is configured to engage with the door latch **116** and retain the door latch **116** in the unlatched position based on the post **104** being inserted into the socket **114**. For instance, as the post **104** is inserted into the socket **114**, the tapered portion **124** can push and rotate the door latch **116** from the latched position to the unlatched position, and thereafter retain the door latch **116** in the unlatched position by preventing rotation back to the latched position so long as the post **104** remains in the socket **114**.

(35) The recessed portion **126** includes a recess **128** configured to receive the post latch **120** based on the post **104** being inserted into the socket **114** and the flip door **118** is in the raised position. For instance, once the post **104** has been inserted into the socket **114** and the flip door **118** is raised such that the dog **122** disengages from the post latch **120**, the post latch **120**—which is aligned with the recessed portion **126**—can rotate into the recess **128**, thereby preventing the post **104** from being removed from the socket **114**.

(36) To facilitate this in some embodiments, the post latch **120** can have a locking portion **130** and a cam portion **132**. With this arrangement, the dog **122** of the flip door **118** is configured to engage with the cam portion **132** and retain the post latch **120** in the unlocked position based on the flip door **118** being in the lowered position. Further, based on the post **104** being inserted into the socket **114** of the safety latch mechanism **108** and the flip door **118** being raised to the raised position, the dog **122** is configured to disengage from the cam portion **132**, thereby causing the post latch **120** to rotate from the unlocked position to the locked position, whereupon the locking

portion **130** is received into the recess **128** of the post dog **106** and locks the post **104** in the safety latch mechanism **108**.

(37) In some embodiments, the body **112** of the safety latch mechanism **108** has a top portion **134**, a bottom portion **136**, and a pair of wall portions **138** that connect the top portion **134** to the bottom portion **136**. In such a scenario, the door latch **116** is rotatably coupled to one of the pair of wall portions **138** (e.g., on an interior surface of one of the walls) and the post latch **120** is rotatably coupled to the top portion **134** (e.g., on an underside of the top portion **134** or on a top surface of the top portion **134**, opposite the underside).

(38) In some embodiments, the safety latch mechanism **108** is configured to be coupled to the fixed structure **110** between the flip door **118** and a second flip door **140** that is adjacent to the flip door **118**. In such a scenario, the door latch **116** and the post latch **120** are rotatably coupled to a first side **142** of the body **112**, and the safety latch mechanism **108** also includes a second door latch **144** rotatably coupled to a second side **146** of the body **112**, opposite the first side **142**, and biased in the latched position that is configured to retain the second flip door **140** in the lowered position. The safety latch mechanism **108** yet further includes a second post latch **148** rotatably coupled to the second side **146** of the body **112** and biased in the locked position. Additionally, the second flip door **140** can be configured similarly to the flip door **118**, and can thus be rotatably coupled to the fixed structure **110** such that the second flip door **140** can rotate between the lowered position and the raised position. As such, the second flip door **140** includes a second dog **150** that is configured to engage with and retain the second post latch **148** in an unlocked position based on the second flip door **140** being in the lowered position.

(39) Yet further in such a scenario, the system **100** can also include an intermediate flip arm **152** rotatably coupled to the safety latch mechanism **108**. The intermediate flip arm **152** can be configured to be mounted between the flip door **118** and the second flip door **140**, and also configured to move between the lowered position and the raised position. Additionally, the post **104** can include a toe **154** that is configured to engage with a top surface **156** of the intermediate flip arm **152** based on the post **104** being inserted into the socket **114** of the safety latch mechanism **108** and retain the intermediate flip arm **152** in the lowered position.

(40) In some embodiments, the fixed structure **110** has a walking surface **158** and a support bracket **160**. The support bracket **160** has an interior surface **162** facing a space underneath the walking surface **158**, as well as an exterior surface **164** that is opposite the interior surface **162** and that faces away from the space underneath the walking surface **158**. In such a scenario, the safety latch mechanism **108** is coupled to the exterior surface **164** of the support bracket **160**. That is, the safety latch mechanism **108** is outboard-facing and coupled to a face of the fixed structure **110**. Further, in such a scenario, the system **100** can also include the flip door **118** and the fixed structure **110**.

(41) FIG. 2 depicts the safety latch mechanism **108** of the system **100**, according to an example implementation. As shown, the post latch **120** is rotatably coupled to the top portion **134** of the body **112** by way of a pivot pin screw **166** defining an axis of rotation that is substantially perpendicular to the top portion **134** of the body **112**. Although not explicitly shown in FIG. 2, the second post latch **148** on the second side **146** of the body **112** can be rotatably coupled in a similar way to the top portion **134** of the body **112**.

(42) As further shown, the door latch **116** is rotatably coupled to one of the pair of wall portions **138** on the first side **142** of the body **112** by way of a pivot pin **168**, and the second door latch **144** is rotatably coupled to the same one of the pair of wall portions **138** on the second side **146** of the body **112** by way of a pivot pin **170**. Pivot pin **168** and pivot pin **170** each define a respective axis of rotation that is substantially perpendicular to the pair of wall portions **138**.

(43) FIG. 3 and FIG. 4 each depict the guardrail **102**, according to an example implementation. FIG. 3 depicts the guardrail **102** having two posts **172**, with each post having the configuration of the post **104** and the post dog **106** described above. As shown, the guardrail **102** includes a foldable top rail **174**, as well as two extending rails **176** that are perpendicular to the foldable top rail **174**.

FIG. 3 specifically depicts the guardrail **102** in a partially-folded scenario. Some embodiments of the guardrail **102** can be foldable for storage purposes, and can be unfolded before insertion into sockets of safety latch mechanisms. In alternative embodiments, the guardrail **102** can include only a single post, configured in the same way or a similar way as the post **104** described above.

(44) FIG. 4 depicts a zoomed-in view of the post **104**, showing the tapered portion **124**, the recessed portion **126**, and the recess **128** of the post dog **106** described above.

(45) FIGS. 5-15 depict various operating scenarios involving the system **100**, as illustrated and described in reference to FIG. 1. For clarity, only a subset of the total structures of the system **100** are specifically numbered in FIGS. 5-15.

(46) FIG. 5 depicts an operating scenario in which a plurality of flip doors are in the lowered position. As shown, the plurality of flip doors include the flip door **118** and two other flip doors that are adjacent to the flip door **118**—namely, the second flip door **140** and a third flip door **178**. Four safety latch mechanisms coupled to the exterior surface **164** of the support bracket **160** and four intermediate flip arms are shown as well, including two safety latch mechanisms **180** on opposing sides of the flip door **118** (of which the safety latch mechanism **108** is labeled and is representative), and further including two intermediate flip arms **182** (of which the intermediate flip arm **152** is labeled and is representative).

(47) Each of the two safety latch mechanisms **180** is shown to be coupled to the fixed structure **110** between the flip door **118** and a respective flip door of the two other flip doors. Further, each of the flip doors is shown to be rotatably coupled to the fixed structure **110** by way of a hinge. Yet further, each intermediate flip arm is shown to be rotatably coupled to a respective one of the safety latch mechanisms.

(48) In the lowered position, the flip doors and intermediate flip arms extend substantially parallel along a reference plane that is parallel to the walking surface **158** of the fixed structure **110**. Further, the top surfaces of the intermediate flip arms are substantially flush with respect to the top surfaces of the flip doors.

(49) FIG. 6 depicts another view of the operating scenario in which the flip door **118**, the second flip door **140**, and the third flip door **178** are in the lowered position. FIG. 6 also shows the two safety latch mechanisms **180**, each with two respective door latches in the latched position and two respective post latches retained in the unlocked position by way of the dogs of the corresponding flip doors (e.g., the dog **122** of the flip door **118** and the second dog **150** of the second flip door **140**).

(50) As further shown, the flip door **118** is shown to include a strike flange portion **184** and a walking portion **186**, where the strike flange portion **184** is substantially perpendicular to the walking portion **186**, and is also substantially parallel (and adjacent to) the exterior surface **164** when the flip door **118** is in the lowered position. The strike flange portion **184** also includes two flaps on opposing sides of the flip door **118**, although could be comprised of more or less flaps in alternative embodiments. Each of the other two flip doors are shown to be similarly configured, each with a respective strike flange portion and a respective walking portion. With the flip door **118** in the lowered position, the door latch **116** extends in front of the strike flange portion **184**, thus preventing movement of the strike flange portion **184** past the door latch **116**, thereby retaining the flip door **118** in the lowered position.

(51) FIG. 7 depicts another view of the operating scenario of FIG. 6.

(52) FIG. 8 depicts an operating scenario in which the guardrail **102** shown in FIGS. 3 and 4 is inserted into the respective sockets of the two safety latch mechanisms **180**. In example embodiments, another system **200** can include the guardrail **102** having the two posts **172**, of which the post **104** of system **100** is representative, and can also include the two safety latch mechanisms **180**, of which the safety latch mechanism **108** of system **100** is representative.

(53) FIG. 9 depicts another view of the operating scenario in which the two posts **172** of the guardrail **102** are inserted into the respective sockets of the two safety latch mechanisms **180**. The

two safety latch mechanisms **180** are each shown with two respective door latches in the unlatched position, although since the flip door **118** and the second flip door **140** are in the lowered position, the respective post latches of those flip doors are still retained in the unlocked position by way of the dog **122** and the second dog **150**. Because the door latches are each in the unlatched position, the strike flange portion **184** is clear to move past the door latches and the flip door **118** can be moved to the raised position.

(54) FIG. **10** depicts an operating scenario in which the two posts **172** of the guardrail **102** are inserted into the respective sockets of the two safety latch mechanisms **180** and the flip door **118** has been raised to the raised position, but where the second flip door **140** is still in the lowered position.

(55) FIG. **11** depicts another view of the operating scenario in which the flip door **118** has been raised to the raised position. FIG. **11** also shows (via an arrow) a direction of rotation of the post latch **120** from the unlocked position to the locked position that occurs when the dog **122** has disengaged from the cam portion **132** of the post latch **120**. Further, upon rotation, the locking portion **130** is received into the recess **128** of the post dog **106**, thereby locking the post **104** in the safety latch mechanism **108**.

(56) FIG. **12** depicts an operating scenario in which the flip door **118** has been lowered back down to the lowered position, thus engaging the dog **122** with the cam portion **132**. The engagement of the dog **122** with the cam portion **132** rotates (shown via an arrow) the locking portion **130** out of the recess **128**, thereby allowing the post **104** to be removed from the safety latch mechanism **108**.

(57) FIG. **13** again depicts the operating scenario in which the guardrail **102** is inserted and the flip door **118** is in the raised position. As further shown, the toe **154** of the post **104** contacts the top surface **156** of the intermediate flip arm **152**.

(58) FIG. **14** next depicts a zoomed-in view of the operating scenario of FIG. **13**.

(59) FIG. **15** depicts an operating scenario in which, instead of the guardrail **102** being inserted into the two safety latch mechanisms **180**, a pair of guardrails **188** are inserted into the safety latch mechanisms. That is, FIG. **15** depicts a scenario in which posts from two separate guardrails can be used to raise a single flip door, as opposed to two posts from a single guardrail are used to raise a single flip door. As further shown, each of the flip doors, as well as each of the intermediate flip arms, are in the raised position.

(60) FIG. **16** shows a flowchart of an example of a method **300** that could be used with the system **100**, the other system **200**, and elements thereof shown in FIGS. **1-15**. Method **300** may include one or more operations, functions, or actions as illustrated by one or more of blocks **302-306**.

(61) At block **302**, the method **300** includes positioning each of two posts of a guardrail proximate to a respective socket disposed in a respective one of two safety latch mechanisms coupled to a fixed structure on opposing sides of a flip door, each post having a respective post dog and each safety latch mechanism comprising a respective body having the respective socket disposed therein, a respective door latch rotatably coupled to the body and biased in a latched position that retains the flip door in a lowered position, and a respective post latch that is rotatably coupled to the body, biased in a locked position, and retained in an unlocked position via engagement with the flip door based on the flip door being in the lowered position.

(62) At block **304**, the method **300** includes inserting each respective post of the two posts into the respective socket such that the respective post dog engages with and rotates the respective door latch from the latched position to an unlatched position, thereby allowing the flip door to be raised from the lowered position to a raised position.

(63) At block **306**, the method **300** includes raising the flip door from the lowered position to the raised position, thereby disengaging the flip door from the respective post latch, whereupon the respective post latch rotates from the unlocked position to the locked position, engages with the respective post dog, and retains the respective post in the safety latch mechanism.

(64) Within examples, the method **300** can also include lowering the flip door from the raised

position to the lowered position, thereby engaging the flip door with the respective post latch, causing the respective post latch to rotate from the locked position to the unlocked position and disengage from the respective post dog, thereby allowing the respective post to be removed from the respective socket. The method **300** can also include removing each respective post of the two posts from the respective socket, thereby causing the respective door latch to rotate to the latched position and retain the flip door in the lowered position.

(65) Different examples of the system(s), device(s), and method(s) disclosed herein include a variety of components, features, and functionalities. It should be understood that the various examples of the system(s), device(s), and method(s) disclosed herein may include any of the components, features, and functionalities of any of the other examples of the system(s), device(s), and method(s) disclosed herein in any combination or any sub-combination, and all of such possibilities are intended to be within the scope of the disclosure.

(66) The description of the different advantageous arrangements has been presented for purposes of illustration and description, and is not intended to be exhaustive or limited to the examples in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different advantageous examples may describe different advantages as compared to other advantageous examples. The example or examples selected are chosen and described in order to best explain the principles of the examples, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various examples with various modifications as are suited to the particular use contemplated.

Claims

1. A system for preventing a flip door from being raised without a guardrail installed, the system comprising: a guardrail comprising a post, the post having a post dog; and a safety latch mechanism configured to be coupled to a fixed structure, the safety latch mechanism comprising: a body, a socket disposed in the body and configured to receive the post, and a door latch rotatably coupled to the body and biased in a latched position that is configured to retain a flip door in a lowered position, wherein based on the post being inserted into the socket of the safety latch mechanism, the post dog engages with and rotates the door latch from the latched position to an unlatched position, thereby allowing the flip door to be raised from the lowered position to a raised position.
2. The system of claim 1, wherein: the safety latch mechanism further comprises a post latch rotatably coupled to the body and biased in a locked position, the flip door comprises a dog that is configured to engage with and retain the post latch in an unlocked position based on the flip door being in the lowered position, and based on the post being inserted into the socket of the safety latch mechanism and the flip door being raised to the raised position, the dog is configured to disengage from the post latch, thereby causing the post latch to rotate from the unlocked position to the locked position, whereupon the post latch engages with the post dog and retains the post in the safety latch mechanism.
3. The system of claim 2, wherein: based on the flip door being lowered from the raised position to the lowered position, the dog is configured to engage with the post latch, causing the post latch to rotate from the locked position to the unlocked position and disengage from the post dog, thereby allowing the post to be removed from the socket, and based on the post being removed from the socket, the door latch rotates from the unlatched position to the latched position, thereby retaining the flip door in the lowered position.
4. The system of claim 2, wherein: the post dog comprises a tapered portion and a recessed portion, the tapered portion is configured to engage with the door latch and retain the door latch in the unlatched position based on the post being inserted into the socket, and the recessed portion comprises a recess configured to receive the post latch based on the post being inserted into the

socket and the flip door is in the raised position.

5. The system of claim 4, wherein: the post latch comprises a locking portion and a cam portion, the dog is configured to engage with the cam portion and retain the post latch in the unlocked position based on the flip door being in the lowered position, and based on the post being inserted into the socket of the safety latch mechanism and the flip door being raised to the raised position, the dog is configured to disengage from the cam portion, thereby causing the post latch to rotate from the unlocked position to the locked position, whereupon the locking portion is received into the recess of the post dog and locks the post in the safety latch mechanism.

6. The system of claim 2, wherein: the body of the safety latch mechanism comprises a top portion, a bottom portion, and a pair of wall portions that connect the top portion to the bottom portion, the door latch is rotatably coupled to one of the pair of wall portions, the post latch is rotatably coupled to the top portion.

7. The system of claim 2, wherein: the safety latch mechanism is configured to be coupled to the fixed structure between the flip door and a second flip door that is adjacent to the flip door, the door latch and the post latch are rotatably coupled to a first side of the body, the safety latch mechanism further comprises: a second door latch rotatably coupled to a second side of the body, opposite the first side, and biased in the latched position that is configured to retain the second flip door in the lowered position, and a second post latch rotatably coupled to the second side of the body and biased in the locked position, and the second flip door comprises a second dog that is configured to engage with and retain the second post latch in an unlocked position based on the second flip door being in the lowered position.

8. The system of claim 7, further comprising: an intermediate flip arm rotatably coupled to the safety latch mechanism, configured to be mounted between the flip door and the second flip door, and configured to move between the lowered position and the raised position, wherein the post comprises a toe configured to engage with a top surface of the intermediate flip arm based on the post being inserted into the socket of the safety latch mechanism and retain the intermediate flip arm in the lowered position.

9. The system of claim 1, further comprising: the flip door; and the fixed structure, wherein: the fixed structure comprises a walking surface and a support bracket, the support bracket having an interior surface facing a space underneath the walking surface, and an exterior surface, opposite the interior surface, facing away from the space, and the safety latch mechanism is coupled to the exterior surface of the support bracket.

10. A system for preventing a flip door from being raised without a guardrail installed, the system comprising: a guardrail comprising two posts, each post having a respective post dog; and two safety latch mechanisms configured to be coupled to a fixed structure on opposing sides of a flip door, each safety latch mechanism comprising: a body, a socket disposed in the body and configured to receive a respective post of the two posts, and a door latch rotatably coupled to the body and biased in a latched position that retains the flip door in a lowered position, wherein based on the respective post being inserted into the socket of the safety latch mechanism, the respective post dog engages with and rotates the door latch from the latched position to an unlatched position, thereby allowing the flip door to be raised from the lowered position to a raised position.

11. The system of claim 10, wherein: each safety latch mechanism further comprises a post latch rotatably coupled to the body and biased in a locked position, the flip door comprises a dog that is configured to engage with and retain the post latch in an unlocked position based on the flip door being in the lowered position, and based on the respective post being inserted into the socket of the safety latch mechanism and the flip door being raised to the raised position, the dog is configured to disengage from the post latch, thereby causing the post latch to rotate from the unlocked position to the locked position, whereupon the post latch engages with the respective post dog and retains the respective post in the safety latch mechanism.

12. The system of claim 11, wherein: based on the flip door being lowered from the raised position

to the lowered position, the dog is configured to engage with the post latch, causing the post latch to rotate from the locked position to the unlocked position and disengage from the respective post dog, thereby allowing the respective post to be removed from the socket, and based on the respective post being removed from the socket, the door latch rotates from the unlatched position to the latched position, thereby retaining the flip door in the lowered position.

13. The system of claim 11, wherein: the respective post dog comprises a tapered portion and a recessed portion, the tapered portion is configured to engage with the door latch and retain the door latch in the unlatched position based on the respective post being inserted into the socket, and the recessed portion comprises a recess configured to receive the post latch based on the respective post being inserted into the socket and the flip door is in the raised position.

14. The system of claim 13, wherein: the post latch comprises a locking portion and a cam portion, the dog is configured to engage with the cam portion and retain the post latch in the unlocked position based on the flip door being in the lowered position, and based on the respective post being inserted into the socket of the safety latch mechanism and the flip door being raised to the raised position, the dog is configured to disengage from the cam portion, thereby causing the post latch to rotate from the unlocked position to the locked position, whereupon the locking portion is received into the recess of the respective post dog and locks the respective post in the safety latch mechanism.

15. The system of claim 11, wherein: the body of the safety latch mechanism comprises a top portion, a bottom portion, and a pair of wall portions that connect the top portion to the bottom portion, the door latch is rotatably coupled to one of the pair of wall portions, the post latch is rotatably coupled to the top portion.

16. The system of claim 11, wherein: each safety latch mechanism is configured to be coupled to the fixed structure between the flip door and a respective flip door of two other flip doors that are adjacent to the flip door, the door latch and the post latch are rotatably coupled to a first side of the body, each safety latch mechanism further comprises: a second door latch rotatably coupled to a second side of the body, opposite the first side, and biased in the latched position that is configured to retain the respective flip door in the lowered position, and a second post latch rotatably coupled to the second side of the body and biased in the locked position, and the respective flip door comprises a second dog that is configured to engage with and retain the second post latch in an unlocked position based on the respective flip door being in the lowered position.

17. The system of claim 16, further comprising: two intermediate flip arms, wherein each intermediate flip arm is rotatably coupled to a respective one of the two safety latch mechanisms, configured to be mounted between the flip door and the respective flip door, and configured to move between the lowered position and the raised position, wherein the respective post comprises a toe configured to engage with a top surface of the intermediate flip arm based on the respective post being inserted into the socket of the safety latch mechanism and retain the intermediate flip arm in the lowered position.

18. The system of claim 10, further comprising: the flip door; and the fixed structure, wherein: the fixed structure comprises a walking surface and a support bracket, the support bracket having an interior surface facing a space underneath the walking surface, and an exterior surface, opposite the interior surface, facing away from the space, and the two safety latch mechanisms are coupled to the exterior surface of the support bracket.

19. A method comprising the system of claim 10, the method further comprising: positioning each of the two posts of the guardrail proximate to a respective socket disposed in a respective one of the two safety latch mechanisms coupled to the fixed structure on the opposing sides of the flip door, each safety latch mechanism further comprising a respective post latch that is rotatably coupled to the body, biased in a locked position, and retained in an unlocked position via engagement with the flip door based on the flip door being in the lowered position; inserting each respective post of the two posts into the respective socket such that the respective post dog engages with and rotates the

respective door latch from the latched position to the unlatched position, thereby allowing the flip door to be raised from the lowered position to the raised position; and raising the flip door from the lowered position to the raised position, thereby disengaging the flip door from the respective post latch, whereupon the respective post latch rotates from the unlocked position to the locked position, engages with the respective post dog, and retains the respective post in the safety latch mechanism.

20. The method of claim 19, further comprising: lowering the flip door from the raised position to the lowered position, thereby engaging the flip door with the respective post latch, causing the respective post latch to rotate from the locked position to the unlocked position and disengage from the respective post dog, thereby allowing the respective post to be removed from the respective socket; and removing each respective post of the two posts from the respective socket, thereby causing the respective door latch to rotate to the latched position and retain the flip door in the lowered position.
